

This lecture covers file operations in Python, focusing on how to open, read, write, and append data to files. The lecture uses the `open()` function, which is built-in and doesn't require importing any libraries.

Key Concepts and Operations:

- **Opening a File:** The `open()` function is used to open files. It takes the file name as the first argument and the mode as the second argument (e.g., `open('test1.txt', 'w')`).
- **File Modes:**
 - `'r'` (Read Mode): Opens the file for reading. If the file doesn't exist, it raises an error.
 - `'w'` (Write Mode): Opens the file for writing. If the file exists, it overwrites the content. If the file doesn't exist, it creates a new file.
 - `'a'` (Append Mode): Opens the file for appending. It adds new data to the end of the file without overwriting the existing content. If the file doesn't exist, it creates a new file.
- **Writing to a File:** The `write()` method is used to write data to a file (e.g., `f.write('My name is Sudh')`). The data is not immediately written to the file until the file is closed.
- **Closing a File:** The `close()` method is crucial to save the changes made to the file (e.g., `f.close()`). Data written to the file is only saved after the file is closed.
- **Reading from a File:** The `read()` method is used to read the content of a file (e.g., `file.read()`). It reads the entire file content as a single string.
- **Reading Line by Line:** Iterating through the file object allows reading the file line by line (e.g., `for i in f:`). The `strip()` method can be used to remove newline characters from each line.
- **Reading a Specific Number of Characters:** The `read(n)` method reads a specified number of characters from the file (e.g., `file.read(10)`).
- **Writing Multiple Lines:** The `writelines()` method writes a list of strings to a file. Each string in the list represents a line (e.g., `f.writelines(list_of_lines)`).
- **Reading Multiple Lines:** The `readlines()` method reads all lines from a file and returns them as a list of strings.

Learning Objectives and Takeaways:

- Understand how to open files in different modes (`read`, `write`, `append`).
- Learn how to write data to a file and the importance of closing the file.
- Learn how to read data from a file, both as a whole and line by line.
- Understand how to append data to a file without overwriting existing content.
- Learn how to use `writelines()` to write multiple lines to a file and `readlines()` to read multiple lines from a file into a list.
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